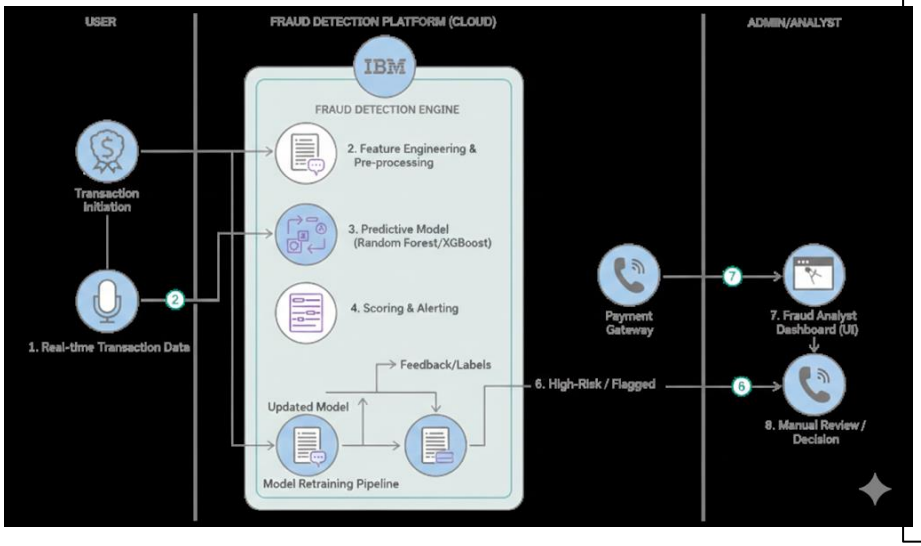
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	LTVIP2026TMIDS80982
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Startup Success Prediction using Machine Learning



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1: Components & Technology

S.No	Component	Description	Technology
1	User Interface	Web interface where users enter startup details and view prediction results	HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	Handles user input validation and request processing	Python, Flask
3	Application Logic-2	Startup prediction processing logic	Scikit-learn, Pandas, NumPy
4	Application Logic-3	Probability calculation and result display	Flask Backend
5	Database	Stores user inputs and prediction history	MySQL
6	Cloud Database	Cloud-based storage for scalability (optional)	Firebase / MySQL Cloud
7	File Storage	Stores trained machine learning model (.pkl file)	Local File System
8	External API-1	Email notification service for prediction results (optional)	SMTP / Gmail API
9	External API-2	User authentication / verification service (optional)	Firebase Auth / Twilio API
10	Machine Learning Model	Predicts whether a startup will be Acquired or Closed	Random Forest / XGBoost
11	Infrastructure (Server / Cloud)	Deployment of application on local or cloud server	Local Server / AWS / Render

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Open-source tools used for development	Flask, Scikit-learn, Bootstrap
2	Security Implementations	Secure authentication, encrypted communication, session control	SHA-256, HTTPS, Flask-Login
3	Scalable Architecture	3-Tier architecture allowing scaling of frontend, backend and database independently	Flask + MySQL + Cloud Hosting
4	Availability	Application accessible 24/7 using cloud infrastructure	AWS / Render Cloud
5	Performance	Real-time startup success prediction with optimized ML model and minimal response time	Optimized ML Model, Caching